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28/9/24

PH3102 Quantum Mechanics **Mid-semester Examination**

Instructor: Dr. Siddhartha Lal **Autumn Semester, 2024**

Instructions: Answer all questions for full marks. All necessary formulae are given. Hand in question paper together with your answer script. Put your name and roll no. on both Question paper as well as the Answer script.

Date: September 28, 2024 **Full marks: 20** **Time: 1.5 hours**

1 Angular Momentum Eigenstates. [6 marks]

Given two angular momentum $\vec{J}_1$ and $\vec{J}_2$ such that $j_1 = 1$ and $j_2 = 1/2$ (where j_α ($\alpha = 1, 2$) denotes the eigenvalue of the J_α^2 operator as $j_\alpha(j_\alpha + 1)\hbar^2$), compute the eigenstates of $\vec{J}_{tot} = \vec{J}_1 + \vec{J}_2$ (i) for the sector $j_{tot} = 3/2$, and (ii) for the sector $j_{tot} = 1/2$. Show all calculations explicitly.
Hint: You may find the following relation useful:

$$J_\pm |j, m_j\rangle \;=\; \hbar\sqrt{j(j+1) - m_j(m_j \pm 1)}\,|j, m_j \pm 1\rangle \ .$$

2 Larmor precession in the Heisenberg picture. [4 marks]

Consider the Hamiltonian H of a particle of spin $\vec{S}$ placed in a magnetic field: $H = BS_z$, where S_z is the spin operator in the $z-$direction. By starting from the respective Heisenberg equations of motion, calculate the rate of change dS_i/dt ($i \in (x, y, z)$) for all three spin operators S_x, S_y and S_z. Given that the initial values of these spin operators at time $t = 0$ are $S_i(0)$ ($i \in (x, y, z)$), solve all three equations to obtain the spin operators $S_i(t)$ as a function of time.

3 The quantum virial theorem. [6 marks]

For a one-dimensional system described by the Hamiltonian $H = \frac{p^2}{2m} + a|x|^n$, where $a > 0$ and $n > -2$, the Virial theorem states that in any eigenstate of H, the following relations are satisfied by the kinetic energy operator $\hat{T}$ and potential energy operator $\hat{V}$

$$\langle\hat{T}\rangle = \langle\frac{p^2}{2m}\rangle = \frac{n}{n+2}\langle H\rangle \ , \ \langle\hat{V}\rangle = \langle a|x|^n\rangle = \frac{2}{n+2}\langle H\rangle \ , \ 2\langle\hat{T}\rangle = n\langle\hat{V}\rangle \ . \tag{1}$$

(a) Consider the Hermitian operator $\hat{D} = [\hat{x}, \hat{p}_x]_+ / 2\hbar = (\hat{x}\hat{p}_x + \hat{p}_x\hat{x})/2\hbar$ (i.e., $[\ ]_+ \equiv$ the anti-commutator of $\hat{x}$ and $\hat{p}_x$) and compute the commutator $\left[\hat{D}, H\right]$.
(b) Prove that for any Hermitian operator $\hat{\Theta}$, the following expectation value computed using the stationary eigenstates ($|\psi\rangle$) of the Hamiltonian (H) must vanish

$$\langle\psi|\left[\hat{\Theta}, H\right]|\psi\rangle = 0 \ . \tag{2}$$

(c) Use the results of parts (a) and (b) to prove the virial theorem.
(d) What is the outcome from the theorem for (i) the case of $n = 2$ (the simple harmonic oscillator) and (ii) the case of Coulomb inter-particle interactions with $n = -1$?

4 The Hellman-Feynman theorem. [4 marks]

For a Hamiltonian H that is a function of a parameter λ, $H \equiv H(\lambda)$, and an eigenstate $|\psi_n\rangle$ with eigenvalue E_n, the Hellmann-Feynman theorem states

$$\frac{\partial E_n}{\partial \lambda} = \langle \psi_n | \frac{\partial H(\lambda)}{\partial \lambda} |\psi_n\rangle \ . \qquad (3)$$

Recall that the harmonic oscillator Hamiltonian $H = \frac{p^2}{2m} + \frac{m\omega^2 x^2}{2}$ is dependent on two parameters, m (mass) and ω (frequency). Use the eigenvalues of the simple harmonic oscialltor ($E_n(\omega)$) and choose the parameter λ carefully so as to implement the Hellmann-Feynman theorem and prove the following for the harmonic oscillator problem:

(a) $\langle \frac{p^2}{2m} \rangle = \langle \frac{m\omega^2 x^2}{2} \rangle$, (b) $\langle \frac{m\omega^2 x^2}{2} \rangle = \frac{1}{2}(n + 1/2)\hbar\omega$.

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